

2nd Year Science – November Exam Revision Worksheet

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Chapter 31 – Speed, Velocity & Acceleration

1. Definitions

a) What is the difference between speed and velocity?

Speed is how fast you are, velocity is speed in a specific direction.

2. Calculations

A car travels 150 km in 3 hours.

(a) Calculate its average speed.

Speed = 50 km/h



$$25 - 10 = 15$$

(b) The same car increases its speed from 10 m/s to 25 m/s in 5 seconds.

Calculate its acceleration.

Acceleration = 3 m/s/s

$$(25 - 10) \div 5$$

$$15 \div 5$$

3. Graphs

The graph below shows a cyclist's journey:

Time (s)	0	2	4	6	8	10
Distance (m)	0	4	8	8	12	16

a) Sketch a **distance-time graph** for this data.

b) Between which times is the cyclist **stationary**?

4-6

c) What kind of motion is shown between 8-10 seconds?

Uniform motion

Chapter 33 – Energy

1. Energy Basics

List four forms of energy:

1. Potential 2. Kinetic 3. Sound 4. Light

2. Law of Conservation of Energy

State this law:

Energy cannot be created or destroyed

3. Energy Conversions

Identify the energy changes in each:

(a) Battery powering a torch →

Chemical energy → Light energy

(b) Toaster →

Electrical energy → heat energy

4. Renewable vs Non-Renewable

Complete the table below:

Type	Example	2 Advantages	2 Disadvantages
Renewable	Wind	Good for environment Can be made anywhere	Costly Expensive
Non-Renewable	Coal	Infrastructure in place Works in all weather	Bad for environment Bad working condition

5. Calculations

- ★ (a) A force of 50 N moves a box 3 m.

Work done = $50 \times 3 = 150$ J

- (b) A light bulb uses 100 J of energy and gives out 25 J as light.

% Efficiency = $\frac{25}{100} \times 100 = 25$ %

6. Sankey Diagram

Draw a simple Sankey diagram for the bulb above.



Chapter 34 – Electric Current

1. Basics

What flows in an electric current? Electrons

Current is measured in amps using an ammeter.

Voltage is measured in voltage using a voltmeter.

2. Circuit Symbols

Draw symbols for:

(a) Cell 	(b) Bulb 	(c) Switch 	(d) Voltmeter
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3. Series vs Parallel

- a) Describe one difference between series and parallel circuits.

If one blows, a parallel circuit can still function

- b) Which type is used in most household circuits? Parallel

4. Conductors & Insulators

Give two examples of each:

Conductor: Water

Insulator: Plastic

5. Circuit Interpretation

Explain what happens when a switch is opened in a series circuit.

It disrupts the current and stops the flow of electrons

Chapter 25 – Structure of the Atom

1. Subatomic Particles

Complete the table:

Particle	Charge	Location
Proton	<i>Positive / +1</i>	<i>Nucleus</i>
Neutron	<i>Neutral / 0</i>	<i>Nucleus</i>
Electron	<i>Negative / -1</i>	<i>Around the nucleus</i>

2. Atomic Number and Mass Number

An atom of sodium has atomic number 11 and mass number 23.

a) Protons = _____ b) Neutrons = _____ c) Electrons = _____

3. Bohr Model

Draw a Bohr model of a sodium atom (Na).

Symbol	Element
H	<i>Hydrogen</i>
O	<i>Oxygen</i>
Na	<i>Sodium</i>
Cl	<i>Chlorine</i>